
APPLIED FUNCTIONAL ANALYSIS
Degree on Applied Mathematics and Computation
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This is a summary of basic concepts discussed in the classroom. The way they are ordered here do not necessarily agree with the way they were presented in the classroom, where the discussion was aimed to motivate and put the ideas in the proper context.

2. Summary: Fundamental notions on linear spaces and linear algebra, normed and Banach spaces.

Definition 1. *Linear spaces. A linear space is a set E together with an addition $+$ and a multiplication by complex or real numbers (scalars) satisfying the following properties:*

1. (Associativity) For any $x, y, z \in E$, $(x + y) + z = x + (y + z)$,
2. (Commutativity) $x + y = y + x$.
3. (Neutral element) For any $x \in E$, there exists an element 0 such that $x + 0 = x$, and
4. (inverse element) there exists another element $-x \in E$ such that $x + (-x) = 0$.
5. For any scalars λ, μ , $\lambda(\mu x) = (\lambda\mu)x$
6. $\lambda(x + y) = \lambda x + \lambda y$, $(\lambda + \mu)x = \lambda x + \mu x$,
7. $1x = x$.

Examples 1. 1. $E = \mathbb{R}^n$ with scalars $\mathbb{R}$; $E = \mathbb{C}^n$ with scalars $\mathbb{C}$.

2. $E = \mathbb{R}[t]$ (polynomials in the variable t) with scalars $\mathbb{R}$.

3. $E = C^0([0, 1])$ with scalars $\mathbb{C}$.

4. $E = L^2([0, 1])$, where $L^2([0, 1]) = \{f: [0, 1] \rightarrow \mathbb{C} \mid \int_0^1 |f(t)|^2 dt < \infty\}$, with scalars $\mathbb{C}$ and $f \sim g$ iff $f(x) = g(x)$, a.e.

Definition 2. Let E be a linear space. A subset $S \subset E$ is said to be a generating system for E if for any $x \in E$, there exists $u_1, \dots, u_r \in S$ and scalars $\lambda_1, \dots, \lambda_r$ such that $x = \sum_{k=1}^r \lambda_k u_k$. If S is a generating system for E it is also said that S generates E .

Definition 3. A subset $S \subset E$ of the linear space E is said to be linearly independent (or free) if for any finite set $u_1, \dots, u_r \in S$ such that $\sum_{k=1}^r \lambda_k u_k = 0$ then necessarily $\lambda_1 = \dots = \lambda_r = 0$.

Definition 4. A Hamel basis of the linear space E is a subset $\mathcal{B} \subset E$ which is free and generates E .

Theorem 1. Let E be a linear space, then it has a Hamel basis.

Normed spaces.

Definition 5. A norm on a linear space E is a map $\|\cdot\|: E \rightarrow \mathbb{R}^+$, satisfying the properties:

- (i) $\|x\| = 0$ iff $x = 0$; (ii) $\|\lambda x\| = |\lambda| \|x\|$; (iii) $\|x + y\| \leq \|x\| + \|y\|$.

Definition 6. A linear space E together with a norm $\| \cdot \|$ will be called a normed space.

Examples 2. 1. Let $E = \mathbb{R}$ with the norm $\|x\| = |x|$. $E = \mathbb{C}$ with the norm $\|z\| = |z|$.

2. $E = \mathbb{R}^n$, $\|x\|_p = \sqrt[p]{\sum_{k=1}^n |x_k|^p}$, $p \geq 1$; $E = \mathbb{C}^n$, $\|z\|_p = \sqrt[p]{\sum_{k=1}^n |z_k|^p}$, $p \geq 1$ (this follows from Hölder's inequality).

3. $E = C^0([0, 1])$, $\|f\| = \max_{t \in [0, 1]} |f(t)|$.

4. $E = L^2([0, 1])$, $\|f\|_2^2 = \int_0^1 |f(t)|^2 dt$.

Proposition 1. If $(E, \| \cdot \|)$ is a normed space, the function $d(x, y) = \|x - y\|$ is a metric and (E, d) is a metric space. The metric d is said to be associated to the norm $\| \cdot \|$.

Definition 7. Let E be a normed space. We will say that E is a Banach space if it is complete with respect to the metric defined by the norm.

Examples 3. 1. The spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ with the norm $\| \cdot \|_2$ are Banach spaces.

2. The spaces l^p , $p \geq 1$, with the norm $\| \cdot \|_p$ are Banach spaces.

3. The space $C^0([0, 1])$ with the norm $\| \cdot \|_\infty$ is a Banach space.

4. The spaces $L^p([0, 1])$ with the norm $\| \cdot \|_p$, $p \geq 1$, are Banach spaces.

This follows from Minkowski inequality: $\|f + g\|_p \leq \|f\|_p + \|g\|_p$, $p \geq 1$, $f, g \in L^p([0, 1])$, which is derived from Hölder's inequality: $\|fg\|_1 \leq \|f\|_p \|g\|_q$, with $1/p + 1/q = 1$.

Definition 8. Two norms $\| \cdot \|_1$, $\| \cdot \|_2$, on the linear space E are equivalent if there exist $0 < A < B$ such that: $A\|x\|_2 \leq \|x\|_1 \leq B\|x\|_2$, for all $x \in E$.

Proposition 2. Two norms on a linear space are equivalent if and only if they possess the same set of convergent sequences.

Proposition 3. If the linear space E is finite dimensional all norms defined on it are equivalent.

Examples 4. The norms $\| \cdot \|_\infty$ and $\| \cdot \|_p$, $p \geq 1$, defined on $C^0([0, 1])$ are not equivalent.